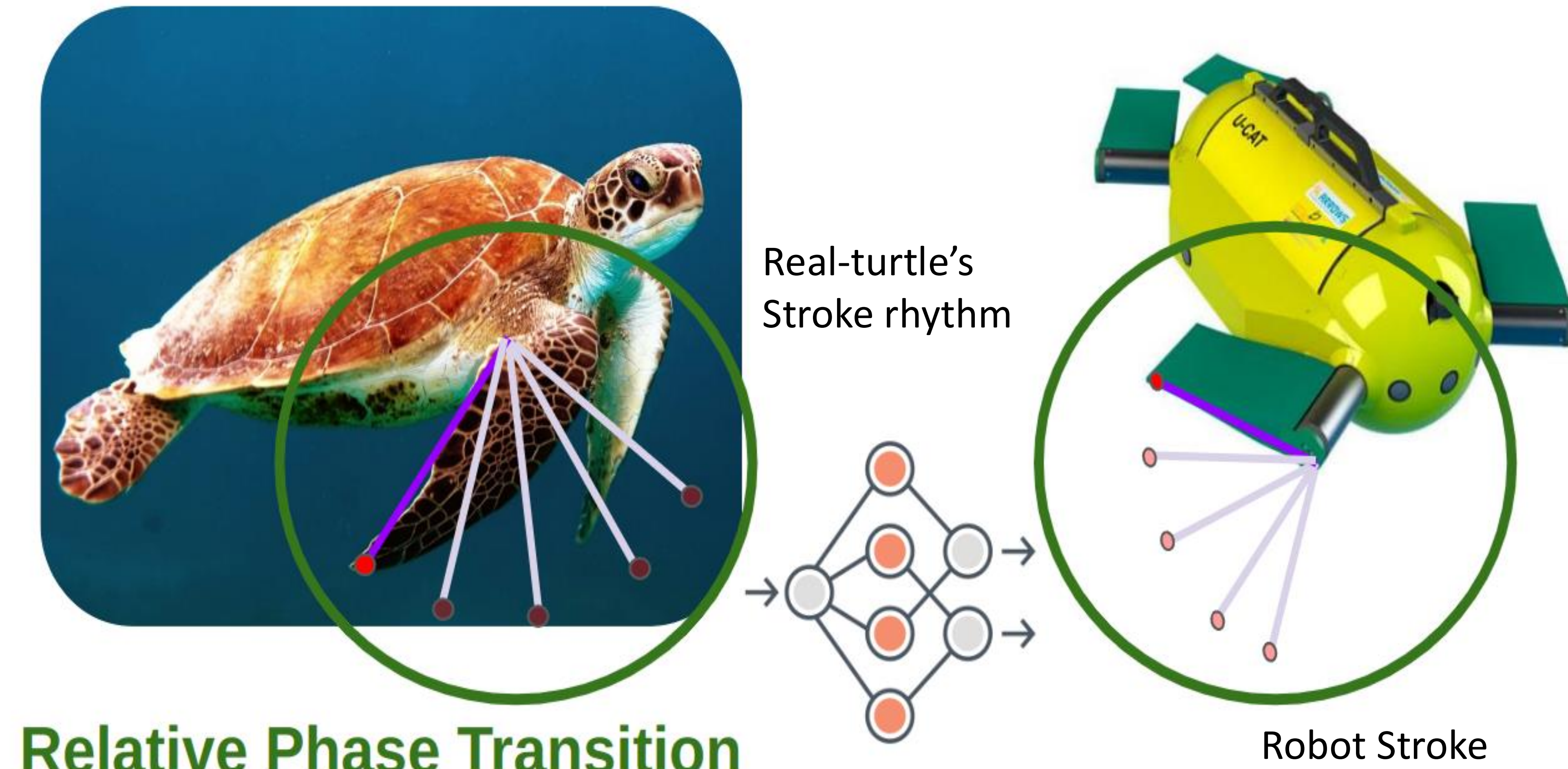
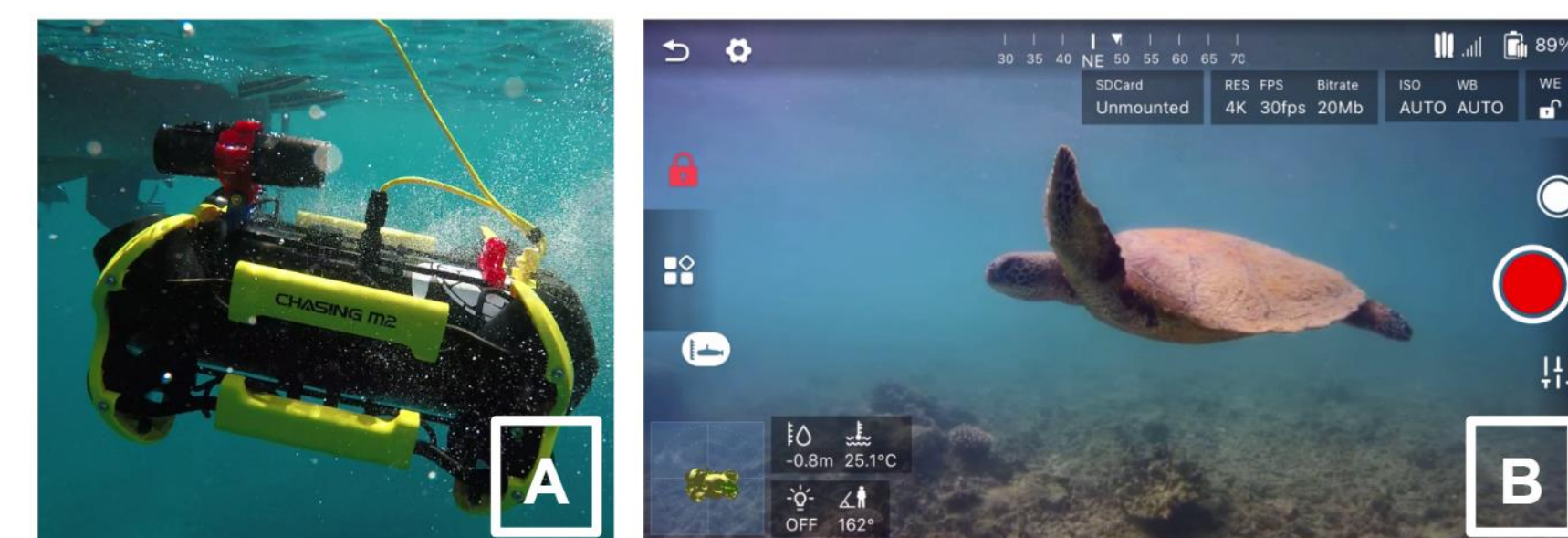


## Overview



## Data collection

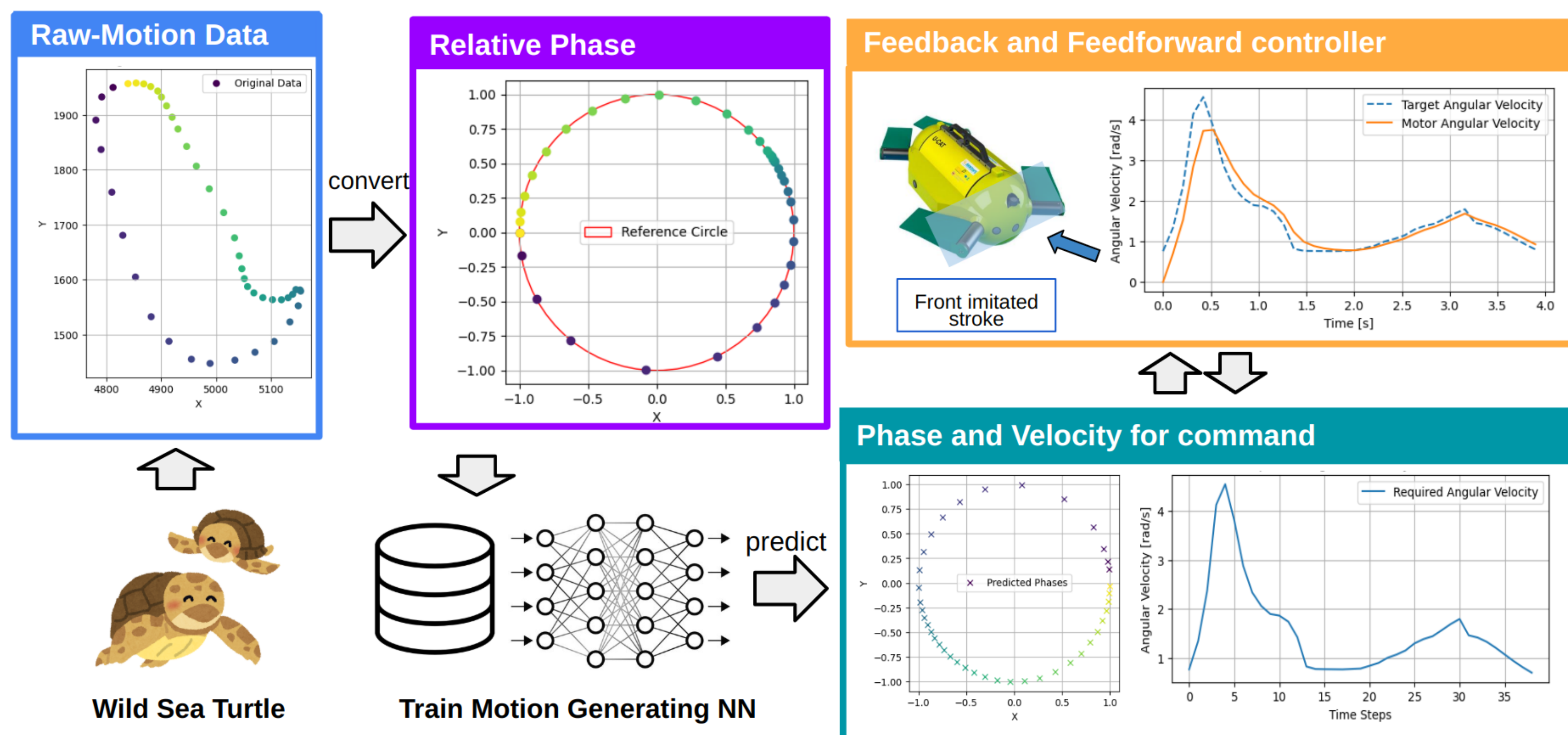


## Adaptation in simulation

$$f(\tau; \mathbf{c}) = c_1 \cos(2\pi\tau) + c_2 \sin(2\pi\tau) + c_3 \cos(4\pi\tau).$$

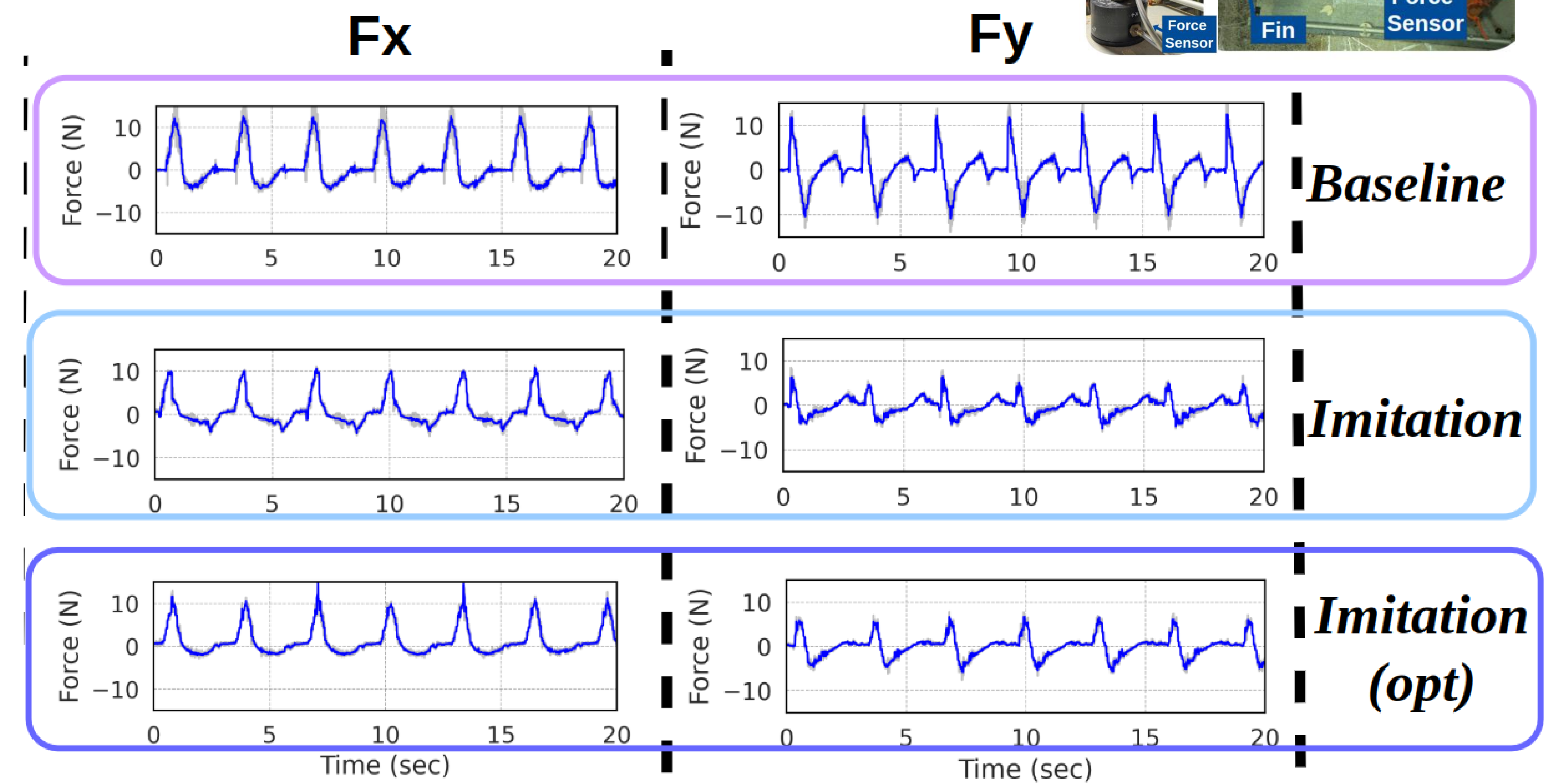
Fine tuning the motion with **envelope** function

## Imitation Pipeline

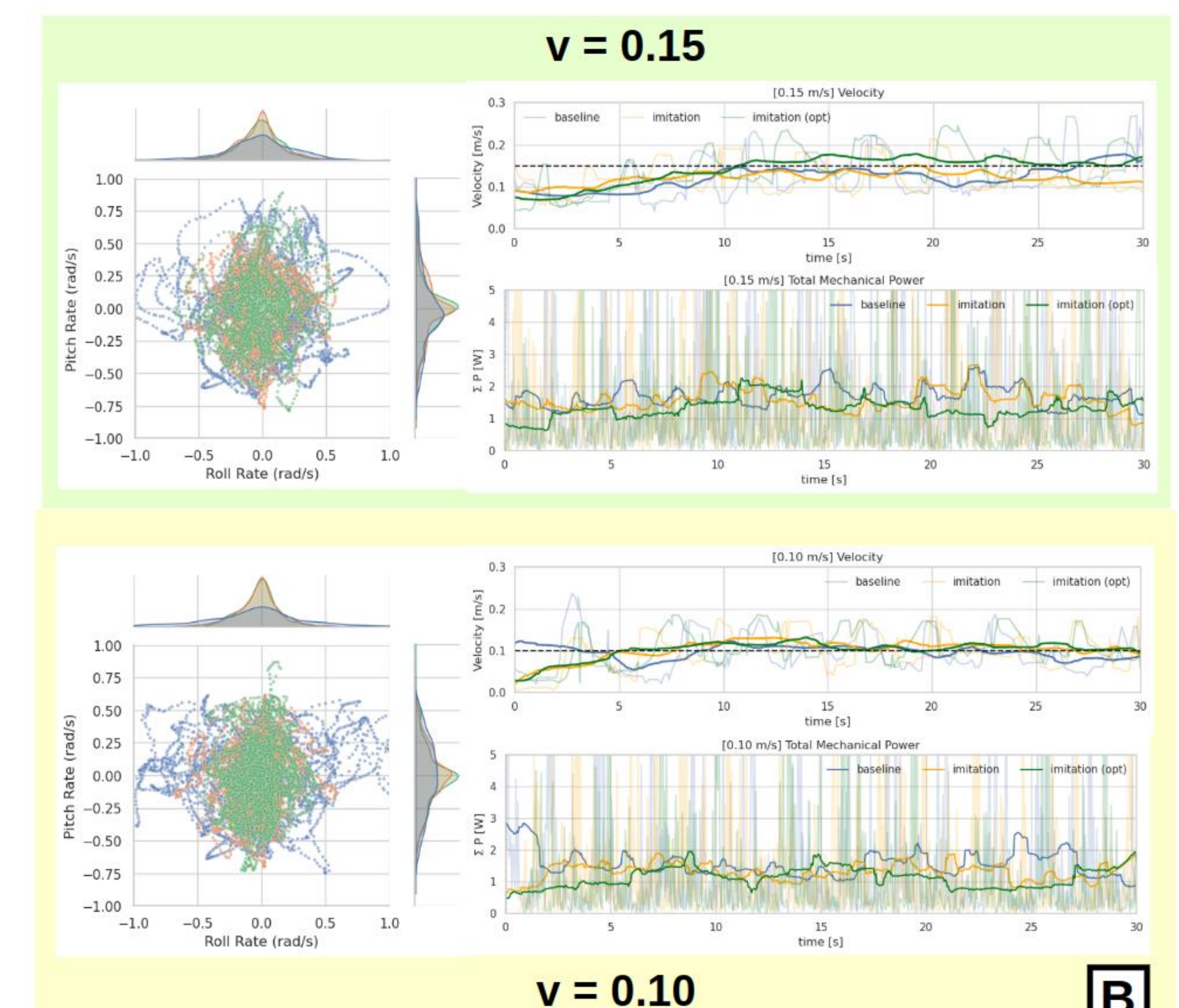
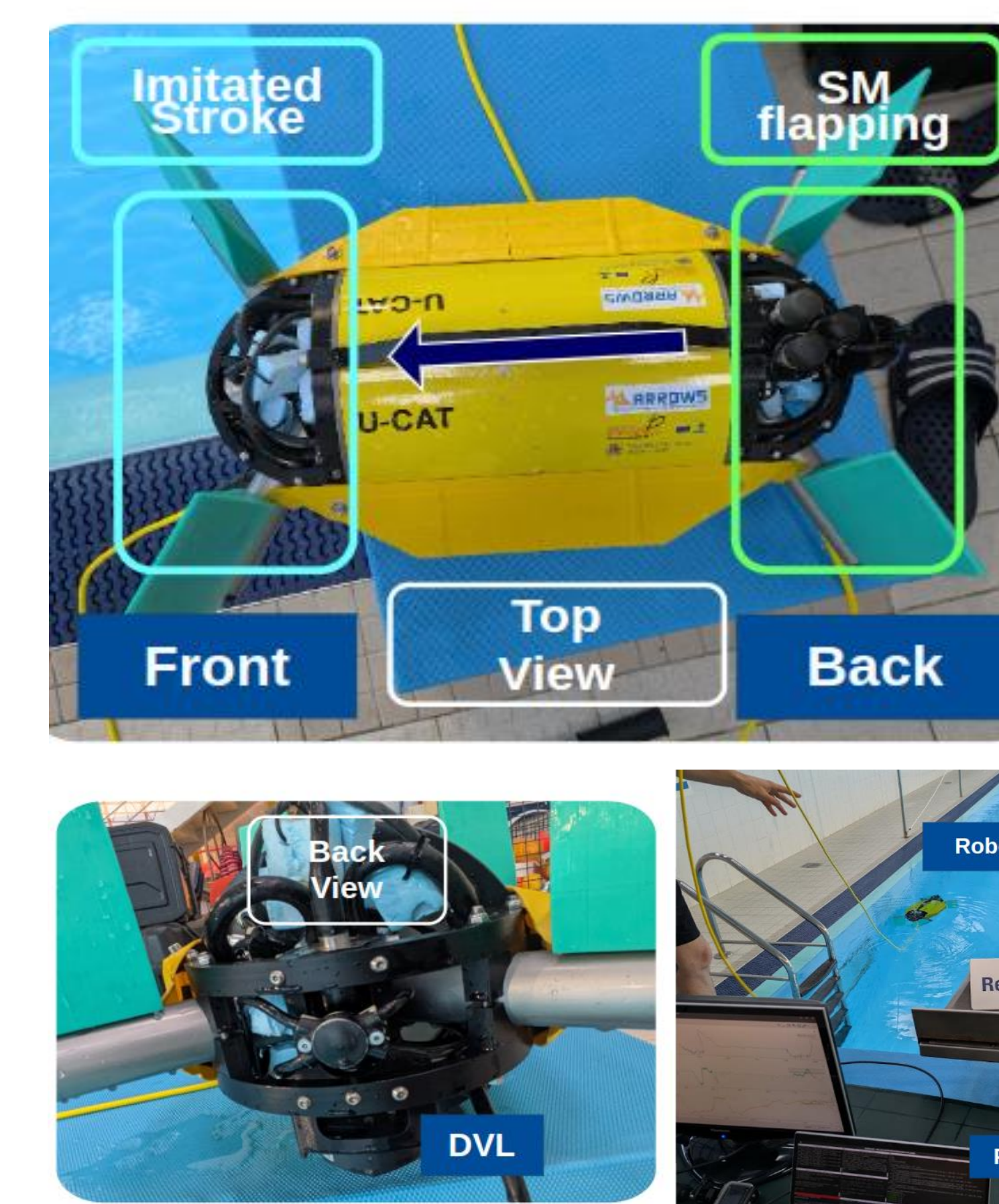


## Experiments

### Single fin results:



### Real robot results:



### Contributions:

- Novel method to design the swimming locomotion from wild animal
- Relative phase transition based imitation learning pipeline
- Adaptation through the simulation and optimization